

Practical - 6

Practical Title: Rendering External JSON Data using Fetch API, Search and Filter

Problem Definition: Display event lists, student profiles, notices, or FAQs from external JSON files using Fetch API. Implement dynamic rendering, search, filter, sorting, and pagination.

Theory :

1. How is JSON fetched, parsed, and rendered?

JSON is fetched using the Fetch API, parsed using `response.json()`, and then rendered dynamically on the webpage using JavaScript DOM methods such as `textContent` or `innerHTML`.

2. Which array methods are used for search, filter, sort, and pagination?

`filter()` is used for search and filtering, `sort()` for sorting, and `slice()` for pagination.

3. How are loading and error states handled?

Loading is handled using the Fetch API, while errors are handled using the `.catch()` method and displayed in the browser console.

4. How is modularity maintained in JavaScript files?

Modularity is maintained by using an external JavaScript file and separating different functionalities into independent code sections or functions.

Data:

```
[
  {
    "id": 1,
    "day": "10",
    "month": "AUG",
    "date": "2026-08-10",
    "title": "Hackathon 2026",
    "location": "Computer Department",
    "time": "10:00 AM",
    "description": "24-Hour Coding Competition with exciting prizes.",
    "category": "Technical",
    "linkText": "Register"
  },
  {
    "id": 2,
    "day": "15",
    "month": "AUG",
    "date": "2026-08-15",
    "title": "Independence Day Celebration",
    "location": "Main Ground",
    "time": "8:00 AM",
    "description": "Flag Hoisting, Cultural Program and Speech.",
    "category": "Cultural",
    "linkText": "View Details"
  },
  {
    "id": 3,
    "day": "20",
    "month": "AUG",
    "date": "2026-08-20",
    "title": "Web Development Workshop",
    "location": "Seminar Hall",
    "time": "11:00 AM",
    "description": "Learn HTML, CSS and JavaScript from industry experts.",
    "category": "Technical",
    "linkText": "Register"
  }
]
```

```
[
  {
    "id": 1,
    "question": "How do I register on StudentHub?",
    "answer": "Click on the <b>Register</b> page, fill in your details, and submit the registration form.",
    "category": "Account"
  },
  {
    "id": 2,
    "question": "How do I login?",
    "answer": "Enter your registered email and password on the Login page.",
    "category": "Account"
  }
]
```

```

1  [
2  { "id": 1, "name": "Aarav Patel", "email": "aarav.patel@example.com", "course": "B.Tech CSE", "year": "2nd Year", "status": "Active" },
3  { "id": 2, "name": "Diya Shah", "email": "diya.shah@example.com", "course": "B.Tech IT", "year": "3rd Year", "status": "Active" },
4  { "id": 3, "name": "Krishna Mehta", "email": "krishna.mehta@example.com", "course": "B.Tech CE", "year": "1st Year", "status": "Pending" },
5  { "id": 4, "name": "Isha Trivedi", "email": "isha.trivedi@example.com", "course": "B.Tech CSE", "year": "4th Year", "status": "Active" },
6  { "id": 5, "name": "Yash Joshi", "email": "yash.joshi@example.com", "course": "B.Tech IT", "year": "2nd Year", "status": "Active" },
7  { "id": 6, "name": "Meera Desai", "email": "meera.desai@example.com", "course": "B.Tech CE", "year": "3rd Year", "status": "Pending" },
8  { "id": 7, "name": "Rohan Parikh", "email": "rohan.parikh@example.com", "course": "B.Tech CSE", "year": "1st Year", "status": "Active" },
9  { "id": 8, "name": "Anaya Pandya", "email": "anaya.pandya@example.com", "course": "B.Tech IT", "year": "4th Year", "status": "Active" },
10 { "id": 9, "name": "Devansh Rana", "email": "devansh.rana@example.com", "course": "B.Tech CE", "year": "2nd Year", "status": "Active" },
11 { "id": 10, "name": "Kavya Bhatt", "email": "kavya.bhatt@example.com", "course": "B.Tech CSE", "year": "3rd Year", "status": "Pending" },
12 { "id": 11, "name": "Aryan Solanki", "email": "aryan.solanki@example.com", "course": "B.Tech IT", "year": "1st Year", "status": "Active" },
13 { "id": 12, "name": "Sanya Chauhan", "email": "sanya.chauhan@example.com", "course": "B.Tech CE", "year": "4th Year", "status": "Active" },
14 { "id": 13, "name": "Vivaan Thakkar", "email": "vivaan.thakkar@example.com", "course": "B.Tech CSE", "year": "2nd Year", "status": "Active" },
15 { "id": 14, "name": "Riya Vora", "email": "riya.vora@example.com", "course": "B.Tech IT", "year": "3rd Year", "status": "Pending" },
16 { "id": 15, "name": "Om Chokshi", "email": "om.chokshi@example.com", "course": "B.Tech CE", "year": "1st Year", "status": "Active" }
17 ]
18

```

Implementation/ Output/Result

All Categories
Date (Earliest First)

10 AUG

Hackathon 2026
Computer Department
10:00 AM
24-Hour Coding Competition with exciting prizes.
Register

15 AUG

Independence Day Celebration
Main Ground
8:00 AM
Flag Hoisting, Cultural Program and Speech.
View Details

20 AUG

Web Development Workshop
Seminar Hall
11:00 AM
Learn HTML, CSS and JavaScript from industry experts.
Register

Sports Tournament
Sports Complex

Placement Training
Auditorium

Teacher's Day Celebration

All Courses
Name (A-Z)

Name	Email	Course	Year	Status
Aarav Patel	aarav.patel@example.com	B.Tech CSE	2nd Year	Active
Anaya Pandya	anaya.pandya@example.com	B.Tech IT	4th Year	Active
Aryan Solanki	aryan.solanki@example.com	B.Tech IT	1st Year	Active
Devansh Rana	devansh.rana@example.com	B.Tech CE	2nd Year	Active
Diya Shah	diya.shah@example.com	B.Tech IT	3rd Year	Active

1
2
3

Course Management

GitHub Repository Link:

<https://github.com/ayushposhiya00-lab/Student-Hub/tree/main/Practical-6>

Learning Outcomes:

- Understand how to fetch and parse external JSON data using the Fetch API.
- Dynamically render JSON data on web pages using JavaScript.
- Implement search, filtering, sorting, and pagination using JavaScript array methods.
- Handle loading and error states while fetching data.
- Develop modular and reusable JavaScript code for dynamic web applications.

Faculty Remarks: (Keep this blank for faculty)